

Scalability & Future Plan

Date	02 July 2026
Team ID	XXXXXX
Project Name	A Comprehensive Measure of Well-Being
Maximum Marks	1 Mark

Scalability & Future Plan:

This document outlines how the current project solution can be scaled to handle larger user bases, increased data loads, or extended features in the future. It also captures the team's roadmap for enhancing and evolving the project beyond its current state.

Current System Limitations:

S.No	Limitation	Impact	Priority to Address
1	Uses a Linear Regression model only	Prediction accuracy may be limited for complex relationships	High
2	Uses a static HDI dataset	Does not reflect the latest socioeconomic data	Medium
3	Runs on a local Flask server	Limited accessibility and concurrent user support	High

Scalability Plan:

S.No	Scalability Aspect	Current State	Proposed Upgrade / Solution
1	User Load	Supports a single local user	Deploy on cloud platforms (Render, Railway, or AWS) to support multiple users.
2	Data Storage	Dataset stored as a local CSV file	Integrate a relational database such as MySQL or PostgreSQL for better data management.
3	Performance	Linear Regression model executed locally	Optimize prediction services and use cloud deployment for faster response times.
4	Security	Basic input validation	Add user authentication, HTTPS, and secure input validation for enhanced security.

Future Roadmap:

Phase	Planned Feature / Enhancement	Target Timeline	Expected Impact
Phase 2	Deploy the application on a cloud platform and integrate a database.	3Months	Improved accessibility and scalability.
Phase 3	Implement advanced machine learning models such as Random Forest or XGBoost for higher prediction accuracy.	6Months	More accurate and reliable HDI predictions.
Phase 4	Add real-time data integration, interactive dashboards, and country-wise comparison features.	9–12 Months	Enhanced analysis, visualization, and decision-making capabilities.